

Sri Lanka Institute of Information Technology



IT2130 – Operating Systems and System Administration
Year 2, Semester 2- 2026

Practical Answer Submission

Practical Sheet No: 09

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Activity 1

1,2

```
dasithi@LAPTOP-88EVVEPR:~$ pwd
/home/dasithi
dasithi@LAPTOP-88EVVEPR:~$ cd/
-bash: cd/: No such file or directory
dasithi@LAPTOP-88EVVEPR:~$ cd
dasithi@LAPTOP-88EVVEPR:~$ ls -l
total 92
-rwxr-xr-x 1 dasithi dasithi 16312 Mar 27 11:07 q1
-rw-r--r-- 1 dasithi dasithi 725 Mar 26 21:18 q1.c
-rwxr-xr-x 1 dasithi dasithi 16472 Mar 27 11:07 q2
-rw-r--r-- 1 dasithi dasithi 814 Mar 26 21:33 q2.c
-rwxr-xr-x 1 dasithi dasithi 16416 Mar 27 11:07 q3
-rw-r--r-- 1 dasithi dasithi 1937 Mar 26 22:11 q3.c
-rw-r--r-- 1 dasithi dasithi 1937 Mar 27 11:30 q3.c.save
-rwxr-xr-x 1 dasithi dasithi 15960 Mar 26 21:05 test
-rw-r--r-- 1 dasithi dasithi 78 Mar 26 21:03 test.c
dasithi@LAPTOP-88EVVEPR:~$
```

Question-How can you recognize file/directory by looking at this output?

Answer- By looking at the First letter:

d → directory

- → file

l → link

Question-Do you have the read, write, or execute permission to these files/directories? What do these fields mean?

Answer- Yes, I have read, write, and execute permissions for the files where I am the owner. From the ls -l output, the first field (such as -rwxr-xr-x) represents the file type and access permissions. The first character indicates whether it is a file (-) or a directory (d). The next nine characters are divided into three groups, representing permissions for the owner, group, and others. For example, rwx means read, write, and execute permissions are granted, while r-x means only read and execute permissions are allowed. The next fields show the number of links, the owner name, the group name, the file size in bytes, the last modified date and time, and finally the file name. Therefore, I as

the owner have full access to my files, while other users have limited permissions depending on the settings.

3.

```
dasithi@LAPTOP-88EVVEPR:~$ cd /bin
dasithi@LAPTOP-88EVVEPR:/bin$ ls -l
total 132616
lrwxrwxrwx 1 root root      4 Feb 10 2024 NF -> coll
lrwxrwxrwx 1 root root      1 Apr  8 2024 X11 -> .
-rwxr-xr-x 1 root root    55744 Jun 22 2025 '['
-rwxr-xr-x 1 root root    18744 Aug 15 2025 aa-enabled
-rwxr-xr-x 1 root root    18744 Aug 15 2025 aa-exec
-rwxr-xr-x 1 root root    18736 Aug 15 2025 aa-features-abi
-rwxr-xr-x 1 root root    16422 Jul  3 2025 add-apt-repository
-rwxr-xr-x 1 root root    14720 Sep 16 2025 addpart
lrwxrwxrwx 1 root root      26 Dec  3 20:31 addr2line -> x86_64-linux-g
nu-addr2line
-rwxr-xr-x 1 root root    2322 Apr 18 2024 apport-bug
-rwxr-xr-x 1 root root   13625 Jul  8 2025 apport-cli
lrwxrwxrwx 1 root root     10 Jul  8 2025 apport-collect -> apport-bu
g
-rwxr-xr-x 1 root root    3790 Jul  8 2025 apport-unpack
-rwxr-xr-x 1 root root  141544 Apr  8 2024 appstreamcli
lrwxrwxrwx 1 root root      6 Apr  8 2024 apropos -> whatis
-rwxr-xr-x 1 root root    18824 Oct 22 2024 apt
lrwxrwxrwx 1 root root      18 Jul  3 2025 apt-add-repository -> add-a
pt-repository
-rwxr-xr-x 1 root root    88544 Oct 22 2024 apt-cache
-rwxr-xr-x 1 root root    27104 Oct 22 2024 apt-cdrom
-rwxr-xr-x 1 root root    31120 Oct 22 2024 apt-config
-rwxr-xr-x 1 root root    23008 Oct 22 2024 apt-extracttemplates
-rwxr-xr-x 1 root root   227816 Oct 22 2024 apt-ftparchive
-rwxr-xr-x 1 root root    51680 Oct 22 2024 apt-get
-rwxr-xr-x 1 root root    28664 Oct 22 2024 apt-key
-rwxr-xr-x 1 root root    55776 Oct 22 2024 apt-mark
-rwxr-xr-x 1 root root    51608 Oct 22 2024 apt-sortpkgs
lrwxrwxrwx 1 root root      19 Dec  3 20:31 ar -> x86_64-linux-gnu-ar
-rwxr-xr-x 1 root root   35336 Jun 22 2025 arch
lrwxrwxrwx 1 root root      19 Dec  3 20:31 as -> x86_64-linux-gnu-as
-rwxr-xr-x 1 root root     984 Jan  2 2024 automat-visualize3
lrwxrwxrwx 1 root root      21 Apr  8 2024 awk -> /etc/alternatives/aw
k
-rwxr-xr-x 1 root root    55816 Jun 22 2025 b2sum
-rwxr-xr-x 1 root root    39432 Jun 22 2025 base32
```

```
dasithi@LAPTOP-88EVVEPR:/bin$ cd /home
dasithi@LAPTOP-88EVVEPR:/home$ ls -l
total 4
drwxr-x--- 5 dasithi dasithi 4096 Mar 27 11:30 dasithi
dasithi@LAPTOP-88EVVEPR:/home$ cd /dev
dasithi@LAPTOP-88EVVEPR:/dev$ ls -l
total 0
crw-r--r-- 1 root root    10, 235 Apr  5 21:19 autofs
drwxr-xr-x 2 root root    600 Apr  5 21:19 block
drwxr-xr-x 2 root root    120 Apr  5 21:19 bsg
crw----- 1 root root    10, 234 Apr  5 21:19 btrfs-control
drwxr-xr-x 2 root root   2860 Apr  5 21:19 char
crw----- 1 root root      5,  1 Apr  5 21:19 console
lrwxrwxrwx 1 root root    11 Apr  5 21:19 core -> /proc/kcore
crw----- 1 root root    10, 123 Apr  5 21:19 cpu_dma_latency
crw----- 1 root root    10, 203 Apr  5 21:19 cuse
drwxr-xr-x 6 root root    120 Apr  5 21:19 disk
crw-rw-rw- 1 root root    10, 125 Apr  5 21:19 dxg
lrwxrwxrwx 1 root root    13 Apr  5 21:19 fd -> /proc/self/fd
crw-rw-rw- 1 root root      1,  7 Apr  5 21:19 full
crw-rw-rw- 1 root root    10, 229 Apr  5 21:19 fuse
crw----- 1 root root    10, 228 Apr  5 21:19 hpet
drwxr-xr-x 2 root root      0 Apr  5 21:19 hugepages
crw----- 1 root root   229,  0 Apr  5 21:19 hvc0
crw----- 1 root root   229,  1 Apr  5 21:19 hvc1
crw--w---- 1 root tty    229,  2 Apr  5 21:19 hvc2
crw----- 1 root root   229,  3 Apr  5 21:19 hvc3
crw----- 1 root root   229,  4 Apr  5 21:19 hvc4
crw----- 1 root root   229,  5 Apr  5 21:19 hvc5
crw----- 1 root root   229,  6 Apr  5 21:19 hvc6
crw----- 1 root root   229,  7 Apr  5 21:19 hvc7
crw----- 1 root root    10, 183 Apr  5 21:19 hwrng
lrwxrwxrwx 1 root root    12 Apr  5 21:19 initctl -> /run/initctl
crw-r--r-- 1 root root      1, 11 Apr  5 21:19 kmsg
crw-rw---- 1 root kvm    10, 232 Apr  5 21:19 kvm
lrwxrwxrwx 1 root root     28 Apr  5 21:19 log -> /run/systemd/journal/
dev-log
crw-rw---- 1 root disk    10, 237 Apr  5 21:19 loop-control
brw-rw---- 1 root disk      7,  0 Apr  5 21:19 loop0
brw-rw---- 1 root disk      7,  1 Apr  5 21:19 loop1
```

```

dasithi@LAPTOP-88EVVEPR:/dev$ cd /etc
dasithi@LAPTOP-88EVVEPR:/etc$ ls -l
total 796
drwxr-xr-x 2 root root      4096 Feb 10 06:26 PackageKit
drwxr-xr-x 7 root root      4096 Feb 10 06:26 X11
-rw-r--r-- 1 root root    3444 Jul  5 2023 adduser.conf
drwxr-xr-x 2 root root      4096 Mar 26 21:05 alternatives
drwxr-xr-x 2 root root      4096 Feb 10 06:26 apparmor
drwxr-xr-x 9 root root      4096 Feb 10 06:26 apparmor.d
drwxr-xr-x 3 root root      4096 Feb 10 06:26 apport
drwxr-xr-x 8 root root      4096 Feb 10 06:26 apt
-rw-r--r-- 1 root root    2319 Mar 31 2024 bash.bashrc
-rw-r--r-- 1 root root      45 Jan 25 2020 bash_completion
drwxr-xr-x 2 root root      4096 Feb 10 06:26 bash_completion.d
-rw-r--r-- 1 root root     367 Aug  2 2022 bindresvport.blacklist
drwxr-xr-x 2 root root      4096 Apr 19 2024 binfmt.d
drwxr-xr-x 2 root root      4096 Feb 10 06:26 byobu
drwxr-xr-x 3 root root      4096 Feb 10 06:24 ca-certificates
-rw-r--r-- 1 root root    6288 Feb 10 06:25 ca-certificates.conf
drwxr-xr-x 5 root root      4096 Mar 26 20:55 cloud
drwxr-xr-x 2 root root      4096 Feb 10 06:25 console-setup
drwx----- 2 root root      4096 Apr 19 2024 credstore
drwxr-xr-x 2 root root      4096 Apr 19 2024 credstore.encrypted
drwxr-xr-x 2 root root      4096 Feb 10 06:25 cron.d
drwxr-xr-x 2 root root      4096 Feb 10 06:26 cron.daily
drwxr-xr-x 2 root root      4096 Feb 10 06:24 cron.hourly
drwxr-xr-x 2 root root      4096 Feb 10 06:24 cron.monthly
drwxr-xr-x 2 root root      4096 Feb 10 06:26 cron.weekly
drwxr-xr-x 2 root root      4096 Feb 10 06:24 cron.yearly
-rw-r--r-- 1 root root    1136 Mar 31 2024 crontab
drwxr-xr-x 4 root root      4096 Feb 10 06:24 dbus-1
drwxr-xr-x 3 root root      4096 Feb 10 06:26 dconf
-rw-r--r-- 1 root root    2967 Apr 12 2024 debconf.conf
-rw-r--r-- 1 root root      11 Apr 22 2024 debian_version
drwxr-xr-x 2 root root      4096 Feb 10 06:26 default
-rw-r--r-- 1 root root    1706 Jul  5 2023 deluser.conf
drwxr-xr-x 2 root root      4096 Feb 10 06:25 depmod.d
drwxr-xr-x 3 root root      4096 Feb 10 06:25 dhcp

```

```

dasithi@LAPTOP-88EVVEPR:/etc$ cd /lib
dasithi@LAPTOP-88EVVEPR:/lib$ ls -l
total 280
drwxr-xr-x 2 root root      4096 Apr  8 2024 X11
drwxr-xr-x 2 root root      4096 Feb 10 06:25 apparmor
drwxr-xr-x 5 root root      4096 Feb 10 06:25 apt
drwxr-xr-x 2 root root      4096 Mar 26 21:04 bfd-plugins
drwxr-xr-x 2 root root      4096 Apr  5 22:00 binfmt.d
drwxr-xr-x 3 root root      4096 Feb 10 06:26 byobu
drwxr-xr-x 2 root root      4096 Feb 10 06:26 cloud-init
-rwxr-xr-x 1 root root     1075 Jan 20 2023 cnf-update-db
-rwxr-xr-x 1 root root    3565 Jan 20 2023 command-not-found
drwxr-xr-x 2 root root      4096 Feb 10 06:26 compat-ld
drwxr-xr-x 2 root root      4096 Feb 10 06:24 console-setup
lrwxrwxrwx 1 root root      21 Jan 31 2024 cpp -> /etc/alternatives/cpp
drwxr-xr-x 2 root root      4096 Feb 10 06:25 dbus-1.0
drwxr-xr-x 3 root root      4096 Feb 10 06:25 dhcpcd
drwxr-xr-x 3 root root      4096 Feb 10 06:24 dpkg
drwxr-xr-x 2 root root      4096 Apr  5 22:00 environment.d
drwxr-xr-x 2 root root      4096 Feb 10 06:25 file
drwxr-xr-x 3 root root      4096 Mar 26 21:04 gcc
drwxr-xr-x 2 root root      4096 Feb 10 06:26 girepository-1.0
drwxr-xr-x 3 root root      4096 Feb 10 06:26 git-core
drwxr-xr-x 2 root root      4096 Feb 10 06:26 gnupg
drwxr-xr-x 2 root root      4096 Feb 10 06:26 gnupg2
drwxr-xr-x 2 root root      4096 Feb 10 06:26 gold-ld
drwxr-xr-x 3 root root      4096 Feb 10 06:25 groff
drwxr-xr-x 2 root root      4096 Feb 10 06:24 init
drwxr-xr-x 3 root root      4096 Feb 10 06:25 kernel
drwxr-xr-x 2 root root      4096 Feb 10 06:26 landscape
drwxr-xr-x 3 root root      4096 Mar 30 2024 locale
drwxr-xr-x 3 root root      4096 Feb 10 06:24 lsb
drwxr-xr-x 2 root root      4096 Feb 10 06:25 man-db
drwxr-xr-x 3 root root      4096 Feb  4 2024 mime
drwxr-xr-x 2 root root      4096 Feb 10 06:25 modprobe.d
drwxr-xr-x 3 root root      4096 Mar 26 20:55 modules
drwxr-xr-x 2 root root      4096 Feb 10 06:25 netplan
drwxr-xr-x 8 root root      4096 Feb 10 06:24 networkd-dispatcher
drwxr-xr-x 2 root root      4096 Feb 10 06:25 openssl

```

Special observation in /dev:

- First letter often c or b
 - c -character device
 - b -block device

4.

```
dasithi@LAPTOP-88EVVEPR:/lib$ cd
dasithi@LAPTOP-88EVVEPR:~$ cd ~
dasithi@LAPTOP-88EVVEPR:~$ cd $HOME
dasithi@LAPTOP-88EVVEPR:~$
```

Question-Can you distinguish the difference among cd, cd ., cd .. commands?

Answer- cd - goes to home

cd ~ - home directory

cd \$HOME -same (environment variable)

Activity 2

1.

```
dasithi@LAPTOP-88EVVEPR:~$ mkdir it24103883
dasithi@LAPTOP-88EVVEPR:~$ cd it24103883
dasithi@LAPTOP-88EVVEPR:~/it24103883$ touch chkpermission
dasithi@LAPTOP-88EVVEPR:~/it24103883$ ls -l
total 0
-rw-r--r-- 1 dasithi dasithi 0 Apr  5 22:04 chkpermission
```

2.

```
dasithi@LAPTOP-88EVVEPR:~/it24103883$ chmod 700 chkpermission
dasithi@LAPTOP-88EVVEPR:~/it24103883$ ls -l
total 0
-rwx----- 1 dasithi dasithi 0 Apr  5 22:04 chkpermission
```

3.

```
dasithi@LAPTOP-88EVVEPR:~/it24103883$ chmod 755 chkpermission
dasithi@LAPTOP-88EVVEPR:~/it24103883$ ls -l
total 0
-rwxr-xr-x 1 dasithi dasithi 0 Apr  5 22:04 chkpermission
```

4.

```
dasithi@LAPTOP-88EVVEPR:~/it24103883$ cd ..
dasithi@LAPTOP-88EVVEPR:~$ ls -l
total 96
drwxr-xr-x 2 dasithi dasithi 4096 Apr  5 22:11 it24103883
-rwxr-xr-x 1 dasithi dasithi 16312 Mar 27 11:07 q1
-rw-r--r-- 1 dasithi dasithi  725 Mar 26 21:18 q1.c
-rwxr-xr-x 1 dasithi dasithi 16472 Mar 27 11:07 q2
-rw-r--r-- 1 dasithi dasithi  814 Mar 26 21:33 q2.c
-rwxr-xr-x 1 dasithi dasithi 16416 Mar 27 11:07 q3
-rw-r--r-- 1 dasithi dasithi  1937 Mar 26 22:11 q3.c
-rw-r--r-- 1 dasithi dasithi  1937 Mar 27 11:30 q3.c.save
-rwxr-xr-x 1 dasithi dasithi 15960 Mar 26 21:05 test
-rw-r--r-- 1 dasithi dasithi   78 Mar 26 21:03 test.c
```

Default-drwxr-xr-x

5,6

```
dasithi@LAPTOP-88EVVEPR:~$ chmod o-rwx it24103883
dasithi@LAPTOP-88EVVEPR:~$ chmod u-rx it24103883
dasithi@LAPTOP-88EVVEPR:~$ ls it24103883
ls: cannot open directory 'it24103883': Permission denied
```

7.

```
dasithi@LAPTOP-88EVVEPR:~$ chmod u+r it24103883
dasithi@LAPTOP-88EVVEPR:~$ ls it24103883
ls: cannot access 'it24103883/chkpermission': Permission denied
ls: cannot access 'it24103883/file2': Permission denied
chkpermission file2
```

8.

```
dasithi@LAPTOP-88EVVEPR:~$ chmod u+rx it24103883
dasithi@LAPTOP-88EVVEPR:~$ ls it24103883
chkpermission file2
```

Observation-Now directory contents are visible

Activity 3

1,2

```
dasithi@LAPTOP-88EVVEPR:~$ mkdir ~/lab1
dasithi@LAPTOP-88EVVEPR:~$ cd ~/lab1
dasithi@LAPTOP-88EVVEPR:~/lab1$ nano idioms
dasithi@LAPTOP-88EVVEPR:~/lab1$ ls -l
total 4
-rwxrwxrwx 1 dasithi dasithi 190 Apr  5 22:24 idioms
dasithi@LAPTOP-88EVVEPR:~/lab1$ chmod a+rw idioms
dasithi@LAPTOP-88EVVEPR:~/lab1$ ls -l
total 4
-rw-r--r-- 1 dasithi dasithi 190 Apr  5 22:24 idioms
```

3,4

```
dasithi@LAPTOP-88EVVEPR:~/lab1$ ls -i idioms
10897 idioms
dasithi@LAPTOP-88EVVEPR:~/lab1$ ls -l idioms
-rw-r--r-- 1 dasithi dasithi 190 Apr  5 22:24 idioms
```

Activity 4

1,2,3

```
dasithi@LAPTOP-88EVVEPR:~/linkdemo$ mkdir softln hardln
dasithi@LAPTOP-88EVVEPR:~/linkdemo$ cd hardln
dasithi@LAPTOP-88EVVEPR:~/linkdemo/hardln$ ln ~/lab1/idioms hardidioms
dasithi@LAPTOP-88EVVEPR:~/linkdemo/hardln$ ls -l
total 4
-rw-r--r-- 2 dasithi dasithi 190 Apr  5 22:24 hardidioms
dasithi@LAPTOP-88EVVEPR:~/linkdemo/hardln$ ls -i
10897 hardidioms
```

Observation-

Same inode ,Same size, Link count increases

4.

```
10897 hardidioms
dasithi@LAPTOP-88EVVEPR:~/linkdemo/hardln$ nano ~/lab1/idioms
dasithi@LAPTOP-88EVVEPR:~/linkdemo/hardln$
```

```
dasithi@LAPTOP-88EVVEPR: ~  
GNU nano 7.2 /home/dasithi/lab1/idioms  
Try, Try, Try, One day you can fly  
Ideas without action are worthless  
What doesn't kill me makes me stronger  
Patience is the companion of wisdom  
Only those who do nothing make no mistakes
```

Observation- changes appear in hardidioms too.

5.

```
dasithi@LAPTOP-88EVVEPR:~/linkdemo/hardln$ ln ~/lab1/idioms hardidioms2
```

Observation- same inode and link count increases

```
dasithi@LAPTOP-88EVVEPR:~/linkdemo/hardln$ rm hardidioms2
```

Observation- link count decreases

6,7

```
dasithi@LAPTOP-88EVVEPR:~/linkdemo/hardln$ cd ../softln  
dasithi@LAPTOP-88EVVEPR:~/linkdemo/softln$ ln -s ~/lab1/idioms softidioms  
dasithi@LAPTOP-88EVVEPR:~/linkdemo/softln$ ls -l  
total 0  
lrwxrwxrwx 1 dasithi dasithi 25 Apr  5 22:38 softidioms -> /home/dasithi/lab  
1/idioms  
dasithi@LAPTOP-88EVVEPR:~/linkdemo/softln$ ls -li  
2246 softidioms
```

Observation- different inode, shows- softidioms -> originalfile

9.

Hard Link:

- Same inode
- Same data
- Increases link count

- Works even if original deleted (until all links removed)

Soft Link:

- Different inode
- Just a pointer
- Breaks if original file deleted